Capstone Project Alignment and Digital Portfolio Integration

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MSIT 5910-01: MSIT Capstone Project

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Part A – Alignment of Capstone Project

Course Learning Objectives

My capstone is a transmission-gap detection system for remote patient monitoring. The final version, 0.4.0, is a single Python file that reads vendor transmission logs, compares each device's current silence with its own usual gap, and puts the devices that need a call into a ranked queue for nurses. MSIT 5910 has six course learning outcomes (University of the People, n.d.), and the project meets each one through specific work.

The first outcome description pertains to the student working with IT and computer science principles concerning complex problems. My system is based on the simple rule returning the score of the device equal to the number of hours of silence divided by the standard gap, as well as flagged at 2.0. I built this idea into four modules (ingestion, gap scoring, a queue API, and call logging) using SQLite and Python's http.server, drawing on what I know about data processing, web services, and databases. The second outcome is concerned with designing the solution with the usage of modern tools, which can be seen in my Unit 3 architecture diagram, as well as in my decision to drop pandas and Streamlit once I found the standard library was enough. The third outcome pertains to evaluation of the solution against the project's requirements. In case of 240 synthetic patients, the personal baseline successfully gave flags for all 64 devices without false positives while the fixed approach based on the 48-hour rule resulted in 40 false alarms and missed 8 necessary calls. The system was tested with 46 test cases, which found four defects. After the fixes, 45 tests pass, with one expected failure for the cohort report that is not built yet. The fourth outcome, analyzing user needs, came from my own experience in remote patient monitoring. The nurses need to obtain the short list with the reasoning for every flag produced.

Ethics, the fifth outcome, is part of the design: the project uses synthetic data only, ingestion drops blood pressure and pulse fields before anything is stored, the server shows each nurse only her own program, and call logging supports a fairness check for patients with poor connectivity. The sixth outcome, writing well-supported arguments, shows in my proposal, technical report, and initial capstone report.

Program Learning Objectives

The MSIT program outcomes have a wider scope (University of the People, 2025). The first one concerns the ability of graduates to analyze complex computing problems using the ideas from IT, computer science and business. My project was a combination of software design and project management because the scope, timeline, and exclusions were documented before programming began. The second refers to creating and evaluating solutions using industry practices. I tested at six levels (unit, integration, system, acceptance, regression, and performance), merged changes in Git through pull requests, tagged each release, and kept settings such as the flag threshold in environment variables. The third and fourth outcomes, user needs and ethics, go further than this course. The server-side access authorization is based on my cybersecurity course, and the rationale behind every flag comes from human-computer interaction knowledge. The fifth outcome is demonstrated through my capstone written reports.

Why This Alignment Matters

When a project is aligned properly, it becomes harder to overstate the results. If every requirement corresponds to a test and the outcomes received can be measured simply against a baseline, it becomes possible to check the statements made in the report rather than accept them. The alignment has made me write the limitations down rather than keep them to myself. Devices

that normally send every three days still take a median of 142 hours to be flagged, and usability has not been tested with real nurses. From a professional perspective, the alignment is important because the issue is real. I do see silent devices in my work, and the skills mentioned in the outcomes such as testing, version control, ethics, and clear reporting are what employers want to see.

Part B – Digital Portfolio Integration

Published portfolio: https://anamkazani.github.io/rpm-gap-detection-new/

Deliverables

The portfolio already includes the version 0.4.0 source code with both test suites, the architecture diagram, a changelog, and two PDFs: my initial capstone report and my testing and maintenance paper, which shows the four defects and their fixes. Later this unit, the final capstone report will replace the initial one, and I will add a short demo video of the walkthrough and nurse dashboard. Carnegie Mellon University School of Computer Science (2024) recommends screenshots or videos of significant work, so the video matters as much as the code.

Organization and Format

Carnegie Mellon University School of Computer Science (2024) recommends using LinkedIn, GitHub, and a personal site together, and describing each project by its challenge, technologies, your role, and results. The main page is created with the help of GitHub Pages, in compliance with the instructions provided by HoffsTech (2023), with the README working as the main page. The structure is in accordance with what W3Schools (n.d.) proposes for a portfolio and consists of an introduction, project description, and contact information. The page begins with a

short problem description, followed by four modules, results, deliverables, and how to use the system. On LinkedIn, I will provide the link to this page and repositories in the featured section. I will also update my headline according to the field of remote patient monitoring and IT because LinkedIn (2023) explains that a person should choose some strong examples which correspond with his or her personal brand.

Polish, Accessibility, and Technical Proof

The repository is organized into src, tests, design, and docs folders, following the upload process GlobalTechFix (2024) shows. I published the v0.4.0 release, saved the reports as PDFs with clear names, and removed an old draft. When the first version of the architecture diagram did not show on the white page, I replaced it with a black-on-white copy. For privacy, everything uses synthetic data, I removed my student ID from the reports, and I will cut the ID verification segment from my demo video before posting it. For accessibility, the diagram has alt text and GitHub's layout is easy to read on a phone. The run instructions let reviewers run the 46 tests themselves. Finally, I will keep the portfolio updated as the project grows, which the CMU article also stresses.

References

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